

ST 502 - Final Project Report

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1 Abstract

The problem we intend to study is the characteristics of murder in Los Angeles, namely the traits more likely to make you a victim.

We are specifically focusing on cases that are labeled criminal homicide, with crime code 110. The characteristics we are assessing are location of murder (by area code), victim age, victim sex, victim ethnicity, and the number of days between the murder and when the murder was reported.

We apply Maximum Likelihood Estimation (MLE) to fit a Gamma distribution to victim age, deriving estimators for the shape and rate parameters analytically. As a complementary approach, we estimate the same parameters within a Bayesian framework using Metropolis-Hastings MCMC sampling, comparing posterior summaries to MLE results. Hypothesis testing via proportion tests, chi-squared tests, and Fisher's exact test is then used to determine whether victim sex, descent, and geographic area are associated with disproportionate victimization relative to LA demographics.

2 Goodness of Fit

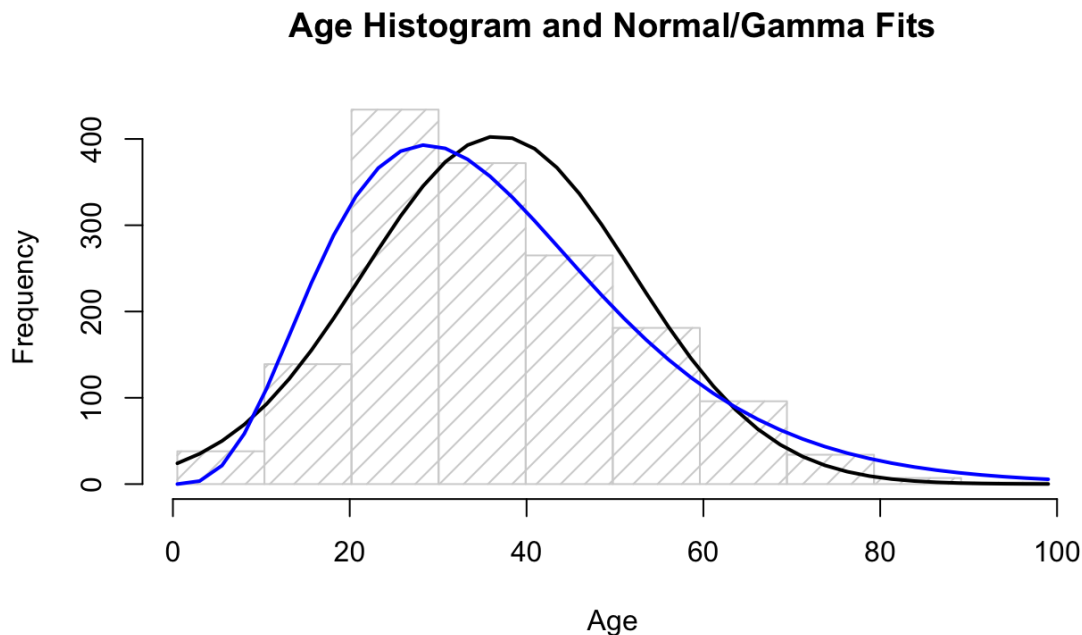


Figure 1: Normal in black, gamma in blue

We fit victim sex to a Bernoulli distribution. Settings values of "M" to 0 and values of "F" to 1, our sample mean (the MLE of a Bernoulli distribution) is 0.1363. Of course, fitting our data to this distribution gives an exact match—the p -value of a χ^2 goodness of fit test is a perfect 1.

To fit victim age, we recognize that many testing procedures assume normal data, so a natural starting point is to check that normality of our data with Pearson's χ^2 goodness of fit test. Starting with age, this test gave a p -value on the order of 10^{-16} for 10 bins, which only decreased as

bins increased, giving ample evidence to reject the null hypothesis that victims' ages are normally distributed.

Next we tried a gamma distribution to help account for the right skew in the data with consistent estimators

$$\hat{\theta} = \overline{x \ln x} - \bar{x} \ln \bar{x}, \quad \hat{\alpha} = \frac{\bar{x}}{\hat{\theta}},$$

but despite the fit visually matching the bins very well, the p -value is still extremely low on the order of 10^{-12} , meaning there is evidence to reject the null hypothesis that victim age follows a gamma distribution.

However, observing that the distribution of victim age is unimodal and right-skewed (Figure 1), we reassessed whether a Gamma distribution truly provides an adequate fit. While a χ^2 goodness of fit test with 10 bins yields a p -value on the order of 10^{-12} , we note that this result should be interpreted cautiously for two reasons. First, with $n = 1568$, the χ^2 test has sufficient power to detect even minor, practically negligible deviations from the assumed distribution. We therefore turn to visual diagnostics as the primary assessment of fit.

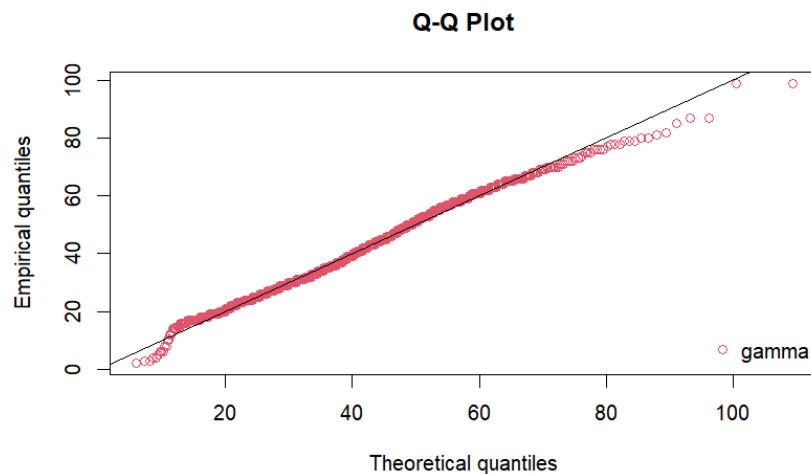


Figure 2: Q-Q plot of victim age against fitted Gamma distribution.

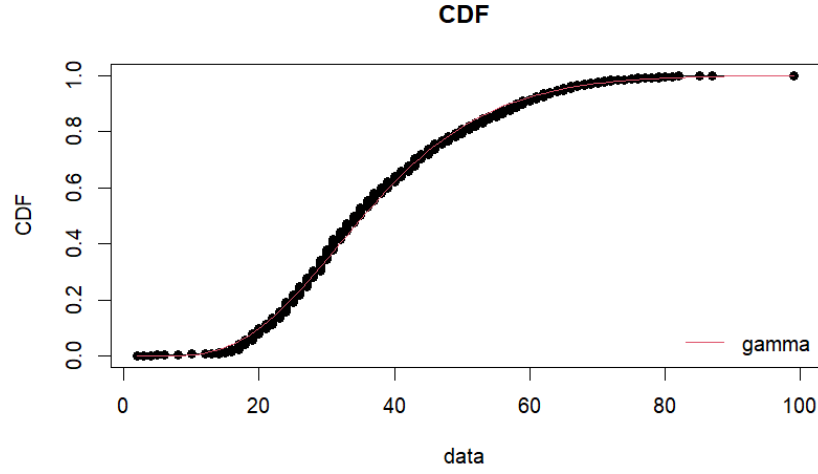


Figure 3: Empirical vs. Theoretical CDF for victim age under fitted Gamma distribution.

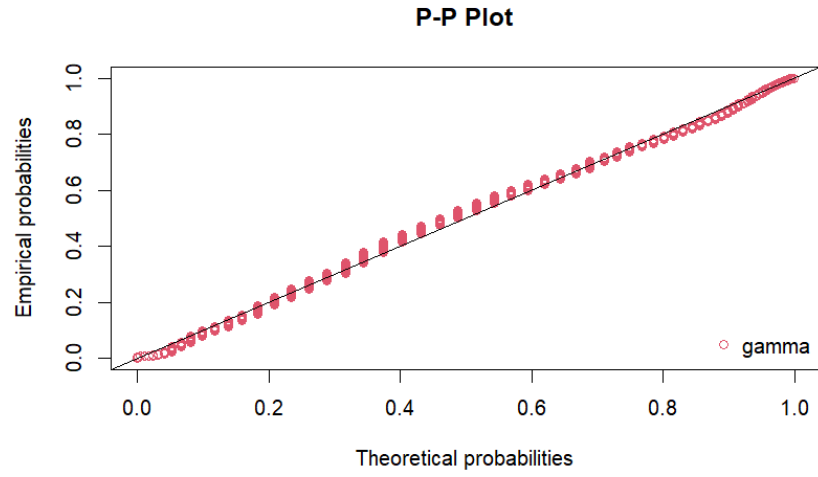


Figure 4: P-P plot of victim age against fitted Gamma distribution.

The Q-Q plot shows close agreement between empirical and theoretical quantiles across the central range of the distribution, with minor departures in the tails: the Gamma slightly underestimates the density of very young victims and underestimates the weight of the upper tail beyond age 80. The CDF and P-P plot both show near-perfect alignment across the full range. On balance, the Gamma distribution provides a reasonable and practically adequate fit to the victim age data, and we proceed under this assumption.

3 Maximum Likelihood Estimation

3.1 Derivation

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$, (under the assumptions that observations are iid) where $\alpha > 0$ is the shape parameter and $\beta > 0$ is the rate parameter, so that the density of each observation is:

$$f(x | \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0.$$

The log-likelihood is:

$$\ell(\alpha, \beta) = n\alpha \log \beta - n \log \Gamma(\alpha) + (\alpha - 1) \sum_{i=1}^n \log x_i - \beta \sum_{i=1}^n x_i.$$

MLE for β . Taking the partial derivative with respect to β and setting it to zero:

$$\frac{\partial \ell}{\partial \beta} = \frac{n\alpha}{\beta} - \sum_{i=1}^n x_i = 0 \implies \hat{\beta} = \frac{\alpha}{\bar{x}}.$$

Since $\hat{\beta}$ depends on α , we substitute and solve for α jointly.

MLE for α . Taking the partial derivative with respect to α :

$$\frac{\partial \ell}{\partial \alpha} = n \log \beta - n\psi(\alpha) + \sum_{i=1}^n \log x_i = 0,$$

where $\psi(\alpha) = \frac{d}{d\alpha} \log \Gamma(\alpha)$ is the digamma function. Substituting $\hat{\beta} = \alpha/\bar{x}$:

$$n \log \left(\frac{\alpha}{\bar{x}} \right) - n\psi(\alpha) + \sum_{i=1}^n \log x_i = 0.$$

This equation has no closed-form solution in α and must be solved numerically. Once $\hat{\alpha}$ is obtained, the MLE for β follows immediately as $\hat{\beta} = \hat{\alpha}/\bar{x}$.

3.2 Results

Fitting the Gamma distribution to the filtered victim age data ($n = 1,561$ after removing records with age coded as zero) yields the maximum likelihood estimates:

$$\hat{\alpha} = 6.3511, \quad \hat{\beta} = 0.1699.$$

The implied mean age under this model is $\hat{\alpha}/\hat{\beta} \approx 37.4$ years, consistent with the sample mean.

4 Bayesian Estimation via Metropolis-Hastings

4.1 Model and Priors

As an alternative to MLE, we estimate α and β within a Bayesian framework. We assign independent weakly informative Gamma priors:

$$\alpha \sim \text{Gamma}(2, 0.1), \quad \beta \sim \text{Gamma}(2, 0.1),$$

chosen to enforce positivity while allowing the likelihood to dominate inference. The joint posterior is:

$$p(\alpha, \beta \mid \mathbf{x}) \propto \mathcal{L}(\alpha, \beta \mid \mathbf{x}) \cdot p(\alpha) \cdot p(\beta),$$

where independence of the priors follows from the assumption that α and β carry no prior dependence. This posterior has no closed form, since the shape parameter α admits no conjugate prior under the Gamma likelihood.

4.2 Metropolis-Hastings Algorithm

We sample from $p(\alpha, \beta \mid \mathbf{x})$ using the Metropolis-Hastings algorithm. At each iteration t :

1. Propose $(\alpha^*, \beta^*) = \exp(\log(\alpha^{(t)}, \beta^{(t)}) + \boldsymbol{\varepsilon})$, where $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$ with $\sigma = 0.03$. Proposing on the log scale ensures positivity of all candidates.
2. Compute the log acceptance ratio:

$$\log r = \log p(\alpha^*, \beta^* \mid \mathbf{x}) - \log p(\alpha^{(t)}, \beta^{(t)} \mid \mathbf{x}).$$

3. Draw $u \sim \text{Uniform}(0, 1)$. Accept the proposal (set $(\alpha^{(t+1)}, \beta^{(t+1)}) = (\alpha^*, \beta^*)$) if $\log u < \log r$, otherwise retain the current value.

The acceptance criterion $\min(1, r)$ is constructed to satisfy detailed balance:

$$p(\theta \mid \mathbf{x}) \cdot T(\theta \rightarrow \theta^*) = p(\theta^* \mid \mathbf{x}) \cdot T(\theta^* \rightarrow \theta),$$

which guarantees that the stationary distribution of the chain is the target posterior. We ran 50,000 iterations initialized at the MLE, discarded the first 10,000 as burn-in, and obtained an acceptance rate of 0.252 — within the conventional target range of 0.2–0.4. Trace plots confirmed rapid mixing and stationarity throughout the retained samples.

4.3 Results

Method	α	β
MLE	6.3511	0.1699
Posterior Mean	6.3553	0.1700
Posterior SD	0.2270	0.0063
Posterior Lower 95% CI	5.9158	0.1578
Posterior Upper 95% CI	6.8065	0.1824

Table 1: MLE estimates and posterior summaries for Gamma parameters fitted to victim age.

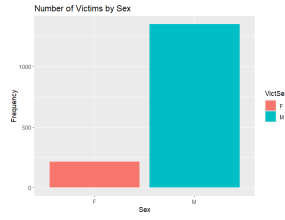
The posterior means are nearly identical to the MLE estimates, which is expected given the large sample size — with $n = 1,561$ observations the likelihood dominates the weakly informative priors and both frameworks converge to the same point estimates. The Bayesian approach additionally provides 95% credible intervals (5.9158, 6.8065) for α and (0.1578, 0.1824) for β . Unlike frequentist confidence intervals, these have a direct probabilistic interpretation: given the observed data, there is a 0.95 posterior probability that each parameter lies within its respective interval. The close agreement between MLE and posterior means was verified to be robust to reasonable changes in the hyperparameters.

5 Hypothesis Testing

To uncover more about what traits are most common in victims, we will conduct a variety of hypothesis tests on our data.

5.1 Gender

First, we'll address the gender divide. According to the 2024 census, women make up 50.2% of the population of Los Angeles. So women should also theoretically make up around 50% of murder victims.



However, in our data, women only represented 213 of the 1566 victims. We will see if this difference is significant via a hypothesis test.

The gender of each murder victim can be represented by a binomial distribution, with n representing the number of victims, and p representing the probability of a victim being female.

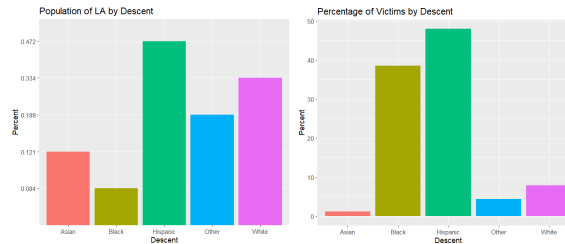
As such, we can test the null hypotheses:

$$H_0 : p = 0.5 \text{ vs. } H_A : p < 0.5$$

using the normal approximation to the binomial for a one-sample proportion test, with a significance level of $\alpha = 0.05$. This test results in a test statistic of 811.18, and a p-value less than $2.2e-16$. Therefore, there is statistically significant evidence that the proportion of murder victims that are female is less than that of men.

5.2 Descents

Next, we'll look at the proportion of various descents of the population of Los Angeles versus our sample.



It's obvious looking at the graph that the murder victims don't line up with the demographics of LA at large. But to find which of these differences are statistically significant, we need to conduct a formal test. For this, we'll use a Chi-Square test, using the proportions of the data vs. the proportions of LA, to test the hypotheses:

$$H_0 : p_A = 0.121, p_B = 0.084, p_H = 0.47, p_O = 0.189, p_W = 0.281$$

vs

$$H_A : \text{at least one not equal}$$

This test results in a test statistic of 164.65, and a p-value less than 2.2e-16. Therefore, there is statistically significant evidence that the population proportions of the victims differ from that of LA at large.

To get more specific with these differences, we'll once again use the normal approximation to the binomial, but this time, p will represent the proportion of the population that belongs to a particular race. We will repeat this for every race. The results of this test are in the table:

Descent	H_A	χ^2	p-value
Asian	Less than	173.48	< 2.2e-16
Black	Greater Than	1848.6	< 2.2e-16
Hispanic	Not Equal	0.32997	0.5657
Other	Less than	213.68	< 2.2e-16
White	Less than	458.27	< 2.2e-16

Table 2: Results of χ^2 Test

The only descent whose victim levels are proportional to the actual demographics was Hispanic/Latino, while Asian, White, and Other have significantly lower proportions, and Black has a significantly higher proportion.

5.3 Precinct Codes

The final categorical predictor is the precinct area the murder occurred in. While it may be tempting to just point to the area code with the largest number of murders, and declare it the most dangerous, but we do need to account for the size of the area. The information on the area can be found here: [link](#). The graphs below show the difference between how the murders would be distributed evenly across the physical area of LA, and the actual distribution.



Once again, it seems that the proportional distribution of cases is not in line with what we would expect. We'll use the same Chi-Square test as last time to solidify this conclusion, with hypotheses:

H_0 : the sample proportions match the expected proportion given a uniform distribution of cases across physical area. H_A : at least one of the proportions doesn't match our expectations

This test results in a test statistic of 3988, and once again a minuscule p-value of less than 2.2e-16. Therefore, we have significant evidence that the true proportion of murders is not evenly spread across the physical area.

Area Code	1	2	3	4	5	6
H_A	GT	GT	GT	GT	LT	NE
Test Stat	470.04	238.39	45.283	174.41	6.0682	2.9848e-31
p-value	< 2.2e-16	< 2.2e-16	8.528e-12	< 2.2e-16	0.006882	>0.999
Area Code	7	8	9	10	11	12
H_A	NE	LT	LT	LT	LT	GT
Test Stat	0.0977	223.49	14.165	60.231	20.839	1097.4
p-value	0.7546	< 2.2e-16	8.372e-05	4.217e-15	2.498e-06	< 2.2e-16
Area Code	13	14	15	16	17	18
H_A	GT	LT	LT	LT	LT	GT
Test Stat	328.83	29.841	8.1364	89.374	116.44	1016.5
p-value	< 2.2e-16	2.345e-08	0.002169	<2.2e-16	<2.2e-16	<2.2e-16
Area Code	19	20	21			
H_A	LT	GT	LT			
Test Stat	16.338	75.291	53.954			
p-value	2.65e-05	< 2.2e-16	1.026e-13			

We'll proceed to use the normal approximation to the binomial to go through the counties one by one to see if they differ from their expected proportion.

Based on these tests, the areas whose proportion of LA murders is higher than it should be are Areas, 1,2,3,4,12,13,18 and 20, which correspond with Central, Rampart, Southwest, Hollenbeck, 77th Street, Newton, Southeast, and Olympic respectively.

The area where the proportion of Murders is less than it should be are areas 5,8,9,10,11,14,15,16,17,19 and 21, which correspond with Harbor, West Los Angeles, Van Nuys, West Valley, Northeast, Pacific, North Hollywood, Foothill, Devonshire, Mission, and Topanga.

The only areas that matched with their expected proportions were Hollywood and Wilshire.

The map below shows the areas color-coded for whether their proportion was more or less than expected.

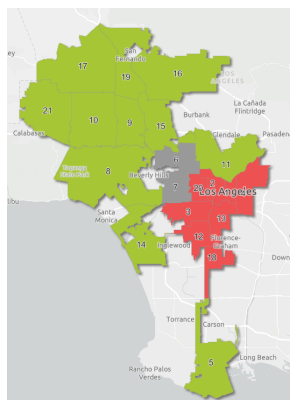


Figure 5: Green areas have a lower proportion, red areas are higher

The visualization of the tests shows that there is an association between geographical location of the precinct and the proportion of murders.

5.4 Difference in Mean Days between Occurrence and Report

Noticing that the vast majority of homicides in Los Angeles are reported on the same day and the vast majority of the remainder are reported on the next day, we can divide cases into those "quickly" reported on the same day and those "slowly" reported on the next day or later. Further separating by gender resolves our data into the table below.

A quick glance shows that about 80% of male victims are reported quickly while only about 75% of female victims are reported quickly. Fisher's exact test shows that this difference is significant, with a p -value of 0.00442.

	Fast	Slow	Total
Male	1095	254	1349
Female	154	59	213
Total	1249	313	1562

Table 3: Gender vs Report Speed

Though Fisher’s exact test will not work for a similar comparison between ethnicity and report speed, Pearson’s χ^2 test for independence will.

	Fast	Slow	Total
Other Asian	12	12	24
Black	495	109	604
Filipino	2	3	5
Hispanic	609	142	751
Other	43	18	61
White	87	36	123
Unknown	4	4	8
Total	1252	324	1576

Table 4: Ethnicity vs Report Speed

This test requires that each category has enough data, for which the slides recommend a rule of thumb of 5. Data for Chinese and Korean victims did not meet this level, so those results were folded into the Other Asian category. A χ^2 test for independence on this data gives a p -value of 6.616×10^{-6} , plenty of evidence to reject the null hypothesis that ethnicity is independent of report time.

6 Conclusions

Through our statistical analysis, we have uncovered several significant characteristics of homicide victims in Los Angeles from 2020 to 2024.

Fitting a Gamma distribution to victim age via MLE yields estimates $\hat{\alpha} = 6.3511$ and $\hat{\beta} = 0.1699$, implying a mean victim age of approximately 37.4 years. Bayesian estimation via Metropolis-Hastings produced posterior means nearly identical to the MLEs, with 95% credible intervals of (5.92, 6.81) for α and (0.158, 0.182) for β . The close agreement between the two frameworks reflects the dominance of the likelihood over the prior at this sample size, and lends consistency to our parameter estimates.

Hypothesis testing revealed that victimization in Los Angeles is far from uniform across demographic groups. Women are significantly underrepresented among homicide victims relative to their share of the population. Black residents are significantly overrepresented, while Asian, White, and Other descent groups are underrepresented; Hispanic victimization is proportional to population share. Geographically, murders are concentrated in central and southern precincts well beyond what their physical area alone would predict. Finally, female victims are reported significantly more slowly than male victims, and reporting speed varies significantly by ethnicity.

Taken together, these findings suggest that the group with the most disproportionately high likelihood of being a homicide victim is Black men residing or spending time in central LA precincts such as Central, Rampart, and 77th Street. The group with the most disproportionately low likelihood are White and Asian women in western and valley precincts. These patterns likely reflect broader socioeconomic and geographic inequalities in Los Angeles rather than individual-level risk factors, and should be interpreted in that context.

7 Appendix: Code

Goodness of fit:

```
g <- Crime_Data$VictAge
g <- replace(g, g==0, 0.5) # Age 0 messes up log: replace with 0.5,
    as the average age of infants between 0 and 1
glogg <- g*log(g)

minAge <- min(g)
maxAge <- max(g)

n <- length(g)
m <- 10 # Number of bins
breaks <- seq(from = minAge+(maxAge-minAge)/m, to = maxAge,
    by = (maxAge-minAge)/m)

theta <- (n*sum(glogg)-sum(g)*sum(log(g)))/n^2
alpha <- mean(g)/theta

h <- hist(g, breaks = append(breaks, 0.5, after = 0), density = 10,
    col = "lightgray", xlab = "Age", main =
    "Age Histogram and Normal/Gamma Fits")
xfit <- seq(min(g), max(g), length = 40)
gamfit <- dgamma(xfit, alpha, scale = theta)
gamfit <- gamfit * diff(h$mids[1:2]) * length(g)
normfit <- dnorm(xfit, mean = mean(g), sd = sd(g))
normfit <- normfit * diff(h$mids[1:2]) * length(g)
lines(xfit, normfit, col = "black", lwd = 2)
lines(xfit, gamfit, col = "blue", lwd = 2)

breakshifts <- breaks
breakshifts[1] <- 0.5
i <- 2
while (i <= m){
    breakshifts[i] <- breaks[i - 1]
    i <- i + 1
}

fbins <- pgamma(breaks, alpha, scale = theta)
    -pgamma(breakshifts, alpha, scale = theta)
tails <- c(pgamma(breakshifts[1], alpha, scale = theta),
    1-pgamma(breaks[m], alpha, scale = theta))

chi2 <- sum((append(h$counts, c(0, 0))-(n*append(fbins, tails)))^2
    /(n*append(fbins, tails)))
df <- m - 1 - 2

p <- pchisq(chi2, df, lower.tail = FALSE)

chisq.test(append(h$counts, c(0, 0)),p=append(fbins, tails))
chi2
```

```
df
p
```

MLE and Bayesian Estimation via Metropolis-Hastings:

```
library(fitdistrplus)
library(tidyverse)

load("Crime_Data.RData")

# Filter out zero ages
ages <- Crime_Data |> filter(VictAge > 0) |> pull(VictAge)

# Precomputed sufficient statistics
n <- length(ages)
sum_x <- sum(ages)
sum_log_x <- sum(log(ages))

# MLE fit
fit_init <- fitdist(ages, "gamma")
cat("MLE shape:", round(fit_init$estimate["shape"], 4), "\n")
cat("MLE rate: ", round(fit_init$estimate["rate"], 4), "\n")

# Goodness of fit diagnostics
denscomp(fit_init, main = "Density")
qqcomp(fit_init, main = "Q-Q Plot")
cdfcomp(fit_init, main = "CDF")
ppcomp(fit_init, main = "P-P Plot")

# Log posterior (unnormalized)
log_posterior <- function(alpha, beta) {
  if (alpha <= 0 || beta <= 0) return(-Inf)
  ll <- n * alpha * log(beta) - n * lgamma(alpha) +
    (alpha - 1) * sum_log_x - beta * sum_x
  lp_alpha <- dgamma(alpha, shape = 2, rate = 0.1, log = TRUE)
  lp_beta <- dgamma(beta, shape = 2, rate = 0.1, log = TRUE)
  ll + lp_alpha + lp_beta
}

# Metropolis-Hastings sampler
mh_sampler <- function(n_iter = 50000, proposal_sd = 0.03) {
  samples <- matrix(NA, nrow = n_iter, ncol = 2)
  colnames(samples) <- c("alpha", "beta")
  current <- c(fit_init$estimate["shape"], fit_init$estimate["rate"])
  accept <- 0
  for (i in 1:n_iter) {
    proposal <- exp(log(current) + rnorm(2, 0, proposal_sd))
    log_ratio <- log_posterior(proposal[1], proposal[2]) -
      log_posterior(current[1], current[2])
    if (log(runif(1)) < log_ratio) {
      current <- proposal
      accept <- accept + 1
    }
  }
}
```

```

    }
    samples[i, ] <- current
  }
  attr(samples, "acceptance_rate") <- accept / n_iter
  samples
}

# Run sampler
set.seed(502)
samples <- mh_sampler()
cat("Acceptance rate:", round(attr(samples, "acceptance_rate"), 3), "\n")

# Burn in
samples_post <- samples[10001:nrow(samples), ]

# Trace plots
par(mfrow = c(2, 1))
plot(samples_post[, "alpha"], type = "l",
      main = "Trace: alpha", ylab = expression(alpha))
plot(samples_post[, "beta"], type = "l",
      main = "Trace: beta", ylab = expression(beta))
par(mfrow = c(1, 1))

# Comparison table
comparison <- data.frame(
  Method = c("MLE", "Posterior Mean", "Posterior SD",
             "Posterior Lower 95%", "Posterior Upper 95%"),
  Alpha = c(fit_init$estimate["shape"],
            mean(samples_post[, "alpha"]),
            sd(samples_post[, "alpha"]),
            quantile(samples_post[, "alpha"], 0.025),
            quantile(samples_post[, "alpha"], 0.975)),
  Beta = c(fit_init$estimate["rate"],
           mean(samples_post[, "beta"]),
           sd(samples_post[, "beta"]),
           quantile(samples_post[, "beta"], 0.025),
           quantile(samples_post[, "beta"], 0.975))
)
rownames(comparison) <- NULL
comparison[, c("Alpha", "Beta")] <- round(comparison[, c("Alpha", "Beta")], 4)
print(comparison)

```

Hypothesis Testing Proportions:

```

library(tidyverse)

data(Crime_Data)

# la demographics from census https://www.census.gov/quickfacts/fact/table/losangelescitycalifornia/
la_demog=data.frame(cbind(Descent=c("A+C+F+K", "B", "H", "O", "W"),
p=c(0.121,0.084,0.472,0.189,0.334)))

```

```

demogp=c(0.121,0.084,0.472,0.189,0.334)

Crime_Data_Clean<-na.omit(Crime_Data)
Crime_Data_Clean$DBR0<-Crime_Data_Clean$'Days Between Report and Occurrence'
Crime_Data_Clean<-Crime_Data_Clean[,-6]
N=nrow(Crime_Data_Clean)
CDC<-Crime_Data_Clean

# is the probability of a murder victim being a women the same as for a man?
# -> NO, p=2.2e-16
Sex<-Crime_Data_Clean|>filter(VictSex!="X")
Sn=nrow(Sex)
prop.test(sum(Sex == "F"), Sn, alternative = "less", p = 0.5)

ggplot(data=Sex,aes(x=VictSex,fill = VictSex))+
  geom_bar()+
  labs(title="Number of Victims by Sex",x="Sex",y="Frequency")

# is the probability of being a murder victim the same for all descents?
# -> NO, p=2.2e-16
Count_Descent=count(Crime_Data_Clean,VictDescent)
Count_Descent$n[1]=Count_Descent$n[1]+Count_Descent$n[3]+Count_Descent$n[4]
+Count_Descent$n[6]
Count_Descent$n[7]=Count_Descent$n[7]+Count_Descent$n[9]
Count_Descent=Count_Descent[-c(3,4,6,9),]
Count_Descent2=Count_Descent
Count_Descent2$n=Count_Descent$n*(100/N)
ggplot(data=Count_Descent2,aes(x=VictDescent,y=n,fill = VictDescent))+
  geom_bar(stat='identity')+
  labs(title="Percentage of Victims by Descent",x="Descent",y="Percent")+
  scale_x_discrete(labels = c(
    "A" = "Asian",
    "B" = "Black",
    "H" = "Hispanic",
    "O" = "Other",
    "W" = "White"))

ggplot(data=la_demog,aes(x=Descent,y=p,fill=Descent))+
  geom_bar(stat='identity')+
  labs(title="Population of LA by Descent",x="Descent",y="Percent")+
  scale_x_discrete(labels = c(
    "A+C+F+K" = "Asian",
    "B" = "Black",
    "H" = "Hispanic",
    "O" = "Other",
    "W" = "White"))

chisq.test(x=Count_Descent2$n,p=demogp)

cup=prop.test(Count_Descent$n[1],N,alternative = "less", p = demogp[1])

```

```

prop.test(Count_Descent$N[2],N,alternative = "greater", p = demogp[2])
prop.test(Count_Descent$N[3],N,alternative = "two.sided", p = demogp[3])
prop.test(Count_Descent$N[4],N,alternative = "less", p = demogp[4])
prop.test(Count_Descent$N[5],N,alternative = "less", p = demogp[5])

# is the probability of being murdered equal across area codes?

Count_Area=data.frame(count(CDC,AreaCode))

print(Count_Area)
Area=cbind(Count_Area,"sarea"=lapddiv$AREA)
Area$sarea=Area$sarea/1000000
Area$ExpProps=Area$sarea/sum(Area$sarea)
Area$RealProps=Area$N/N
Area$ExpMurders=Area$ExpProps*N

ggplot(data=Area,aes(x=AreaCode,y=RealProps,fill=AreaCode))+
  geom_bar(stat='identity')+
  guides(fill="none")+
  labs(x="Area Code", y="Proportion of Cases", title="Real Proportion of
Cases by Area Code")

ggplot(data=Area,aes(x=AreaCode,y=ExpProps,fill=AreaCode))+
  geom_bar(stat='identity')+
  guides(fill="none")+
  labs(x="Area Code", y="Number of Murders", title="Expected Proportion of
Cases by Area Code")

chisq.test(x=Area$N,p=Area$ExpProps)

prop.test(Area$N[1],N,alternative = "greater", p = Area$ExpProps[1])
prop.test(Area$N[2],N,alternative = "greater", p = Area$ExpProps[2])
prop.test(Area$N[3],N,alternative = "greater", p = Area$ExpProps[3])
prop.test(Area$N[4],N,alternative = "greater", p = Area$ExpProps[4])
prop.test(Area$N[5],N,alternative = "less", p = Area$ExpProps[5])
prop.test(Area$N[6],N,alternative = "two.sided", p = Area$ExpProps[6])

prop.test(Area$N[7],N,alternative = "two.sided", p = Area$ExpProps[7])
prop.test(Area$N[8],N,alternative = "less", p = Area$ExpProps[8])
prop.test(Area$N[9],N,alternative = "less", p = Area$ExpProps[9])
prop.test(Area$N[10],N,alternative = "less", p = Area$ExpProps[10])
prop.test(Area$N[11],N,alternative = "less", p = Area$ExpProps[11])
prop.test(Area$N[12],N,alternative = "greater", p = Area$ExpProps[12])

prop.test(Area$N[13],N,alternative = "greater", p = Area$ExpProps[13])
prop.test(Area$N[14],N,alternative = "less", p = Area$ExpProps[14])
prop.test(Area$N[15],N,alternative = "less", p = Area$ExpProps[15])
prop.test(Area$N[16],N,alternative = "less", p = Area$ExpProps[16])
prop.test(Area$N[17],N,alternative = "less", p = Area$ExpProps[17])

```

```
prop.test(Area$n[18],N,alternative = "greater", p = Area$ExpProps[18])

prop.test(Area$n[19],N,alternative = "less", p = Area$ExpProps[19])
prop.test(Area$n[20],N,alternative = "greater", p = Area$ExpProps[20])
prop.test(Area$n[21],N,alternative = "less", p = Area$ExpProps[21])
```

Report speed vs sex:

```
data <- Crime_Data
time <- data$'Days Between Report and Occurrence'
sex <- data$VictSex
```

```
n <- length(time)
```

```
Mnum <- 0
Fnum <- 0
Xnum <- 0
Mfast <- 0
Ffast <- 0
Xfast <- 0
i <- 1
while (i < n){
  if(i == 1158 | i == 1241){
    i <- i + 1
  }
  if (sex[i] == 'M'){
    Mnum <- Mnum + 1
    if (time[i] == 0){
      Mfast <- Mfast + 1
    }
  }
  else if (sex[i] == 'F'){
    Fnum <- Fnum + 1
    if (time[i] == 0){
      Ffast <- Ffast + 1
    }
  }
  else {
    Xnum <- Xnum + 1
    if (time[i] == 0){
      Xfast <- Xfast + 1
    }
  }
  i <- i+1
}
```

```
Mslow <- Mnum-Mfast
Fslow <- Fnum-Ffast
Xslow <- Xnum-Xfast
fast <- Mfast + Ffast + Xfast
```



```

table <- matrix(c(Mfast,Ffast,Mfast+Ffast,Mslow,Fslow,Mslow+Fslow,Mnum,
  Fnum,Mnum+Fnum), ncol = 3, byrow = TRUE)
colnames(table) = c('M','F','Total')
rownames(table) = c('Fast','Slow','Total')
as.table(table)

table2 <- matrix(c(Mfast,Ffast,Mslow,Fslow), ncol = 2, byrow = TRUE)
fisher.test(table2) # p-value 0.004142

```

8 References

- City of Los Angeles Data. (2020, February 10). Crime Data from 2020 to 2024. Lacity.org. [https:// data.lacity.org/Public-Safety/Crime-Data-from-2020-to-2024/2nrs-mtv8/about_data](https://data.lacity.org/Public-Safety/Crime-Data-from-2020-to-2024/2nrs-mtv8/about_data) Julia, P. (2018, February 18).
- LAPD Divisions. Geohub.lacity.org; Los Angeles Geohub. <https://geohub.lacity.org/datasets/lahub::lapd-divisions/about>
- U.S. Census Bureau. (2024, July). QuickFacts: Los Angeles city, California. Census Bureau QuickFacts; United States Census Bureau. <https://www.census.gov/quickfacts/fact/table/losangelescitycalifornia/PST045224>